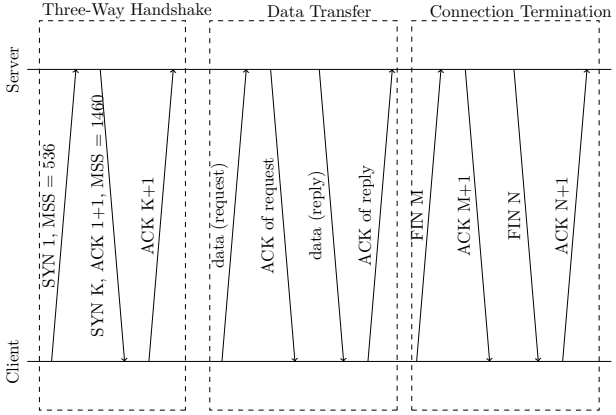
1 Congestion Control

Key terms: Link Capacity R, Congestion Window `cwnd`, N Hosts
Ideal throughput: R/N Measured: $\frac{\#\text{bytes in RTT}}{RTT}$
Cost of Congestion:

- Retransmissions (packet lost)
- Unneeded retransmits (full buffer $\Rightarrow$ 2x send)
- Buffering wasted for packets lost downstream

Types: Loss-based (TCP Reno/Cubic) | Delay-based (Vegas) | ECN (L4S)
TCP uses a **sliding window** via `cwnd`.
TCP Reno: 3 dup ACK $\rightarrow$ SR $= \frac{1}{2}$
TCP Tahoe: timeout $\rightarrow$ SR $= 1$ MSS
Slow-start: cwnd $= 1$, double per round (cwnd++ per ACK); switch to linear after loss; threshold = pre-loss cwnd.



ECN: Sender sets ECN bits (01) in IP header; router marks 11 on congestion; receiver sets ECE bit in ACK.
L4S (Low Latency, Low Loss, Scalable — IPv4+IPv6):

- No drops, only marks $\rightarrow$ congestion signaling
- No starvation; coexists with non-L4S traffic
- Prioritizes L4S; goals: max throughput, min latency
- Prague theorem
- Datarate: $r = \frac{1}{p} - 1$ [Mbps], $p =$ packet marking prob.

Fairness: R/N UDP: constant rate, no CC TCP: allows parallel connections.

2 Flow Control

Goal: Don't overflow receiver buffer.
Link Capacity R: physical interface bandwidth of interconnect cable.
Window-based FC: Max data allowed without ACK (packet loss).
Rate-based FC: Congestion window / sliding window TCP; rate chosen at start.
More speed: Use different algos from CC/FC:
$\rightarrow$ Loss-based vs. Delay-based vs. Hybrid/ECN
"If you can't make the pipe bigger, make the flow smarter"

- Multipath / Load Balancing
- Active Queue Management
- Traffic Compression $\rightarrow$ Better

3 Buffering

n TCP flows $\Rightarrow B = \dfrac{RTT \cdot C}{\sqrt{n}}, \quad C =$ Capacity

4 Congestion vs Flow Control

Congestion $\approx$ too many senders $\neq$ Flow Control $\approx$ one sender too fast.
$\Rightarrow$ Long delays, queueing, packet loss, overflow.

5 Packet Classification

Why: Faster routes | Allow good traffic | Drop malicious
Goals:

- Routing decision $\rightarrow$ select interface / fastest route
- Packet filtering $\rightarrow$ drop / allow
- Accounting $\rightarrow$ measure data
- Rate limiting

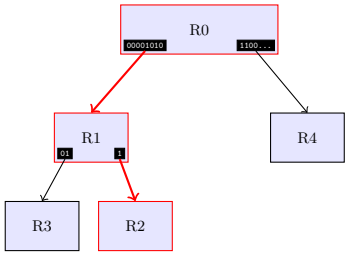
5.1 Binary Representation IPv4

12.4.0.0/15 Mask: 255.254.0.0

Addr	00001100	00000100	00000000	00000000
Mask	11111111	11111110	00000000	00000000

Network Prefix | Hosts

5.2 Unibit Tries



Route P1: `10.128.12.1` = 00001010.1 ... $\rightarrow$ R2
W-bit prefixes, N prefixes: $\mathcal{O}(W)$ lookup, $\mathcal{O}(NW)$ storage, $\mathcal{O}(W)$ updates.

5.3 Bit-Vector Linear Search

Match ruleset: AND bit-vectors per field, pick highest-priority match.
Scenario: Packet `<123.25.1.1, 124, 123.25.4.0, 53, TCP>`

Field	R1	R2	R3	R4	R5	R6	R7	R8
Src-IP	1	1	1	1	0	1	1	1
Src-Port	1	1	1	1	0	1	1	1
Dst-IP	1	1	0	1	0	1	1	1
Dst-Port	0	1	1	0	0	1	1	1
Protocol	1	0	1	1	0	1	1	1
AND	0	0	0	0	0	**1**	**1**	**1**

Hits: R6, R7, R8 $\Rightarrow$ **Rule 6** (highest priority).
Complexity: $O(N \cdot d)$ time & space; with word width W: $O(N/W)$ per field.

6 IPv4 & IPv6

IPv4: 32-bit (10^9) IPv6: 128-bit (10^{38})
Interface: connects to link (virtual/physical)
Netmask: boundary between network & host part
Subnet border: from interface, over links, to all connected interfaces
Router: connects networks $\Rightarrow$ VLAN/subnet traversal
Switch: connects devices in same network ($\rightarrow$ no L3, no IP)
NAT: One IP for all devices in a network
NAT Table: Ext. Address | Ext. Interface | Int. Address | Int. Port
NAT: rewrite IP source $\rightarrow$ Layer 3, NAT Table in router.

IP Count: #Interfaces +2 = Broadcast + Network Address

Private IP Ranges:

0.0.0.0/8	This network
10.0.0.0/8	Local comms in priv. netw.
100.64.0.0/10	Shared Carrier-Grade NAT
127.0.0.0/8	Loopback
169.254.0.0/16	Link-Local
172.16.0.0/12	Local comms in priv. netw.
192.168.0.0/16	Local comms in priv. netw.
224.0.0.0/4	Multicast

Link Local Address = SLAAC (Stateless Address Autoconfiguration)

IPv6 Special Addresses:

`::1/128`	Loopback
`::/128`	Unspecified
`::ffff:X.X.X.X/96`	IPv4 mapped
`64:ff9b::hex(X.X.X.X)/96`	NAT64
`5f00::/16`	IPv6 Segment Routing
`fc00::/7`	Unique Local Address
`fe80::/64`	Link Local Address
`ff00::/8`	Multicast Address
`2000::/3`	Global Unicast
`FC00::/7`	Unique Local Unicast
`fe80::/10`	Link-Local Unicast

7 BGP

AS ↔ AS

Intra: Within same AS **Inter:** Different AS

→ Be careful who you choose (Cost, Privacy)

Tools: AS numbers in path

TCP Port 179

Messages: Open, Update, Keepalive, Notification

Routing Information Protocol → iBGP

Kürzeste Pfade in einem AS finden ⇒ Hop Count

8 Subnets

Subnet = group of interfaces with same prefix.

/N mask ⇒ 2^{32-N} host addresses (IPv4).

Always reserve: network addr. + broadcast addr.

9 Collective Communication

Primitives:

- **Broadcast:** one node sends full vector m to all
- **Scatter:** one node sends chunk m/n to each node
- **Gather:** one node collects chunks from all → full vector
- **AllGather:** every node ends up with full vector (chunks from all)
- **Reduce:** one node collects & applies op (sum/min/max) on index
- **AllReduce** = Reduce-Scatter + AllGather; every node gets result
- **Reduce-Scatter:** each node gets reduction of one coordinate

α-β **Cost Model** (grade Heuristik):

$T_{\text{total}} = \alpha + \beta \cdot m_{\text{size}}$ (α=latency, β=inv. bandwidth)

Butterfly AllGather: $\approx \log_2 n \cdot \alpha + m \cdot \beta$

Memory Footprint **Degree**

9.1 Ring Topology Comparison

	Ring	Butterfly	Swing	Trivance
Latency	Linear	$\log_2 n$	$\log_2 n$	$\log_3 n$
Congestion	None	Exp. grow	Improved	Improved

10 Modelling Topologies as Graphs

Graph $G = (V, E)$: V = set of n nodes/peers, E = edges/links

Distance $d(u, v)$: shortest path from u to v

Diameter $D(G) = \max_{(u,v)\in V^2}\{d(u, v)\}$: longest shortest path

Neighbor set $\Gamma(U) = \{v \in V \setminus U \mid \exists u \in U : (u, v) \in E\}$

= nodes adjacent to U but not part of U

$\alpha(U) = |\Gamma(U)| \setminus |U|$

Expansion $\alpha(G) = \min_{U,\, |U|\le n/2}\{\alpha(U)\}$

→ Expansion covers "bottlenecks"

11 eBPF / RDMA / P4

eBPF: runs programs in privileged context; no kernel modules, no source; runtime for events ⇒ hooks & events.

RDMA = Remote Direct Memory Access

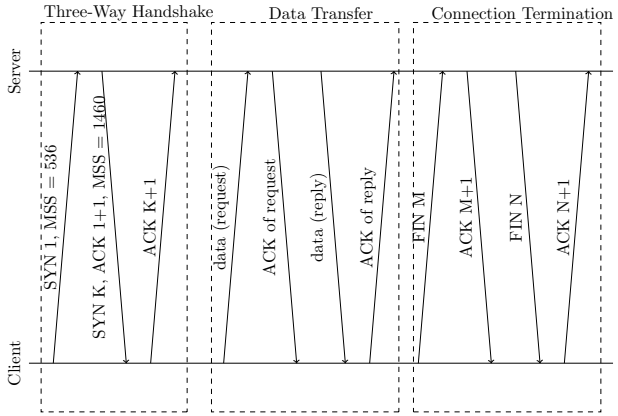
P4: Domain Specific Language

≈ Controls packet forwarding → software

Implementation independent + Protocol independent

No native support for IP/Ethernet/...

12 WIP



13 OSI Model

#	Layer	Protocol/Unit	Key Function
7	Application	HTTP, DNS, FTP	User-facing services
6	Presentation	TLS, SSL	Encoding, encryption
5	Session	Sockets	Session management
4	Transport	TCP, UDP	Segment, port, reliability
3	Network	IP, ICMP, BGP	Packet routing
2	Data Link	Ethernet, MAC	Frame, MAC addr, switches
1	Physical	Cables, WiFi	Bits over medium